RIKATHI PAL

HomePage ripal@cs.stonybrook.edu Google-Scholar Linkedin Github
+1 9342278812 Stony Brook, NY, USA

Research Profile

Computer Science Ph.D. student developing machine learning methods for healthcare, specializing in controllable generative models and medical image analysis. Research includes mask-conditioned 3D diffusion models for anatomy-preserving, pathology-controlled CT synthesis, latent-space medical image generation, and MRI tumor segmentation. My research aligns with the Biomedical Machine Learning Group's interests in medical imaging and generative modeling for healthcare.

Education

- **Stony Brook University, New York, United States** Fall 2025 – Expected 2030
  PhD in Computer Science — Advisor: Prof. Klaus Mueller **GPA: 3.87/4**
  – Research interests: biomedical ML, generative models, and medical imaging.

- **Information Technology, University of Calcutta, India** 2020 – 2024
  B.Tech in Information Technology — Advisors: Amlan Chakrabarti and Somoballi Ghoshal **CGPA: 9.36/10**
  – Research: lumbar spine tumor segmentation in T2-weighted MRI.

Research Experience

Research Assistant **August 2025 – Present**
Stony Brook University, New York, United States
Advisor: Prof. Klaus Mueller
– Developed controllable 3D diffusion models and latent-space methods for medical image generation.
– Conducted quantitative evaluations and downstream classification experiments; collaborated on manuscript preparation.

Teaching Assistant **August 2025 – Present**
Stony Brook University, New York, United States
– Supported day-to-day course operations by grading assignments, maintaining academic records, explaining technical concepts, and mentoring students.
– Communicated technical concepts clearly to diverse audiences through one-on-one support and instructional assistance.

Pre Doctoral Research Fellow **June 2024 – July 2025**
Indian Institute of Science (IISc), Bangalore, India
Studying Neural Network Training via Loss Landscape Analysis Using Topological Data Analysis
– Analyzed neural network training through topological data analysis of loss landscapes and latent representations.
– Applied contour trees and persistent homology to characterize topological structure.
– Collaborated with faculty on experimental evaluation, result interpretation, and research presentations.

Research Intern **May 2023 – Nov 2023**
Indian Statistical Institute (ISI), Kolkata, India
ACEV: Unsupervised Segmentation of Intersecting Manifolds in High-Dimensional Spaces
– Contributed to unsupervised manifold segmentation using k-nearest neighbors and principal component analysis (PCA).
– Investigated adaptive segmentation thresholds based on angular changes of local eigenvectors.
– Conducted experimental evaluations and presented methodological findings to research collaborators.

Medical Imaging Research Projects

Medical Image Generation from 2D Empty Space ***2026–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller
– Generated CT images from user-selected empty regions in 2D projections of a high-dimensional latent space.
– Reconstructed latent vectors using distance-weighted nearest neighbors and cluster-aware selection.
– Evaluated image quality, perceptual similarity, and latent reconstruction accuracy.

Pathology-Controlled Medical Image Generation ***2025–Present***
Stony Brook University, United States — Advisor: Prof. Klaus Mueller
– Developed mask-conditioned 3D DDPMs for anatomy-aware, pathology-controlled lung CT synthesis.
– Designed phase-decoupled global and local guidance to preserve anatomy and control lesion appearance.
– Evaluated synthetic image quality and augmentation for downstream lung lesion classification.

Lumbar Spine Tumor Segmentation & Localization (T2 MRI) ***2023–2024***
B.Tech Research — University of Calcutta, India — Advisor: Amlan Chakrabarti and Somoballi Ghoshal
– Worked on deep learning methods for lumbar spine tumor segmentation and localization in T2-weighted MRI.
– Performed medical image preprocessing and experimental evaluation of segmentation results.
– Co-authored a research paper accepted at NCVPRIPG 2025.

Publications

- ***Under Review*** **- Anatomy-Aware and Pathology-Controlled Diffusion for 3D Lung CT Synthesis**
  Authors: ***Rikathi Pal****, Agransh Srivastava, Arjun Krishna, Klaus Mueller*
  Journal: Submitted to Physics in Medicine and Biology

- **Medical Image Segmentation Using Advanced Attention UNet** **(1st Runner-up)** *DOI No.*
  Authors: ***Rikathi Pal*****, Somoballi Ghoshal, Amlan Chakrabarti*
  Conference: ICDMAI 2025, Date: January 17-19, 2025

- **Lumbar Spine Tumour Segmentation and Localization from T2 MRI Images Using AI** *DOI No.*
  Authors: ***Rikathi Pal****, Somoballi Ghoshal, Piya Saha, Amlan Chakrabarti, Susmita Sur-Kolay*
  Conference: NCVPRIPG 2025, Date: July 19-22, 2025

- **AI Scientist Mission Control (AIMC): Visual Analytics for Human Oversight of Autonomous Scientific Discovery** *DOI No.*
  Authors: ***Rikathi Pal*****, Klaus Mueller*
  Conference: IEEE VIS 2026 Poster, Date: November 9-13, 2026

- **MobileResNet: A Generalized Lightweight Model for Accurate Leaf Disease Detection Across Diverse Plant Types** *DOI No.*
  Authors: ***Rikathi Pal*****, Somoballi Ghoshal, Amlan Chakrabarti*
  Conference: INDICON'24, Date: December 19-21, 2024

- **A Novel Feature Extraction Model for the Detection of Plant Disease from Leaf Images in Low Computational Devices** *DOI No.*
  Authors: ***Rikathi Pal*****, Anik Basu Bhaumik, Arpan Murmu, Sanoar Hossain, Biswajit Maity, Soumya Sen*
  Conference: ICSTA'23, Date: December 19-20, 2023

- ***Under Review*** **- Topological Analysis of DNNs via Loss-Augmented Latent Spaces**
  Authors: ***Rikathi Pal****, Vijay Natarajan, Harish Doraiswamy*
  Journal: Submitted to IEEE Transactions on Visualization and Computer Graphics

Leadership & Volunteer Experience

Student Volunteer (Upcoming) **November 2026**
IEEE VIS 2026, Boston, MA, United States

- Will support conference operations as a student volunteer.

Student Lead – AI/ML Initiatives **April 2023 – March 2024**
Google Developer Student Club (GDSC), University of Calcutta, India

- Led AI/ML team activities, coordinated technical sessions, and tracked collaborative project progress.
- Communicated project methods and outcomes through technical documentation and presentations.

Scholarship & Award

- **Hagan International Scholarship** — Hagan Scholarship Foundation (2025–2027)
- **Second Best Paper Award** — International Conference on Data Management, Analytics & Innovation (ICDMAI 2025)
  Paper: *Medical Image Segmentation Using Advanced Attention UNet*

Technical Skills

- **Programming:** Python, C++, SQL, Java.
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras.
- **Research Methods:** Generative modeling, diffusion models, convolutional neural networks, medical image segmentation, representation learning, unsupervised learning, dimensionality reduction.
- **Scientific Computing and Computer Vision:** NumPy, Pandas, OpenCV, Matplotlib, Seaborn.
- **Development Tools:** Git, Visual Studio Code, LaTeX.